

Numerical Analysis homework 1

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I

$$\begin{aligned} \langle u, v \rangle &= \int_a^b \rho(t) u(t) \overline{v(t)} dx = \int_a^b \rho(t) u(t) \overline{v(t)} dt = \overline{\langle v, u \rangle} \\ \langle \lambda u, v \rangle &= \int_a^b \lambda \rho(t) u(t) \overline{v(t)} dx = \lambda \langle u, v \rangle \\ \langle u, \lambda v \rangle &= \int_a^b \rho(t) u(t) \overline{\lambda v(t)} dx = \bar{\lambda} \langle u, v \rangle \\ \langle u, u \rangle &= \int_a^b \rho(t) |u(t)|^2 dx \geq 0 \end{aligned}$$

II

a

$$\langle t_n, t_m \rangle = \int_{-1}^1 \frac{\cos(n \arccos x) \cos(m \arccos x)}{\sqrt{1-x^2}} dx \quad (1)$$

$$= \int_0^\pi \cos mt \cos nt dx = \begin{cases} 0, & m \neq n \\ \frac{\pi}{2}, & m = n \neq 0 \\ \pi, & m = n = 0 \end{cases} \quad (2)$$

b

Chebyshev: $T_0 = 1, T_1 = x, T_2 = 2x^2 - 1$ normalize: $T_0^* = \sqrt{\frac{1}{\pi}}, T_1^* = \sqrt{\frac{2}{\pi}}x, T_2^* = \sqrt{\frac{2}{\pi}}(2x^2 - 1)$, Chebyshev polynomials are orthogonal on the interval $[-1, 1]$.

III

a

From II.b, we have orthogonal system T_0^*, T_1^*, T_2^* ,

$$\langle y(x), T_0^* \rangle = \int_{-1}^1 \rho(x) y(x) T_0^* dx = \frac{2}{\sqrt{\pi}} \quad (3)$$

$$\langle y(x), T_1^* \rangle = \int_{-1}^1 \rho(x) y(x) T_1^* dx = 0 \quad (4)$$

$$\langle y(x), T_2^* \rangle = \int_{-1}^1 \rho(x) y(x) T_2^* dx = -\frac{2\sqrt{2}}{3\sqrt{\pi}} \quad (5)$$

$$(6)$$

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And Fourier expansion:

$$y(x) = \sum_{n=0}^2 \langle y(x), T_n^* \rangle T_n^* = \frac{2}{\pi} - \frac{4}{3\pi}(2x^2 - 1).$$

b

For normal equation:

$$y(x) = \sum_{n=0}^2 a_n \cdot x^n$$

where a_i are the coefficients to be determined by the equation below.

$$\begin{pmatrix} \langle 1, 1 \rangle & \langle 1, x \rangle & \langle 1, x^2 \rangle \\ \langle x, 1 \rangle & \langle x, x \rangle & \langle x, x^2 \rangle \\ \langle x^2, 1 \rangle & \langle x^2, x \rangle & \langle x^2, x^2 \rangle \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} \langle y(x), 1 \rangle \\ \langle y(x), x \rangle \\ \langle y(x), x^2 \rangle \end{pmatrix}$$

substitute the values,

$$\begin{pmatrix} \pi & 0 & \frac{\pi}{2} \\ 0 & \frac{\pi}{2} & 0 \\ \frac{\pi}{2} & 0 & \frac{3\pi}{8} \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ \frac{2}{3} \end{pmatrix}$$

IV

a

According to the Example 5.55,

$$\langle 1, 1 \rangle = 12, \langle x, x \rangle = 650, \langle x^2, x^2 \rangle = 60710, \quad (7)$$

$$\langle 1, x \rangle = 78, \langle x, x^2 \rangle = 6084, \quad (8)$$

$$\langle 1, x^2 \rangle = 650. \quad (9)$$

Now we Orthonormalize the basis:

$$v_0 = 1 \quad (10)$$

$$u_0^* = \frac{v_0}{\|v_0\|_2} = \frac{1}{\sqrt{12}}; \quad (11)$$

$$v_1 = x - \langle x, u_0^* \rangle u_0^* = x - \frac{78}{12}, \quad (12)$$

$$u_1^* = \frac{v_1}{\|v_1\|_2} = \frac{x - \frac{78}{12}}{\sqrt{650}}; \quad (13)$$

$$v_2 = x^2 - \langle x^2, u_0^* \rangle u_0^* - \langle x^2, u_1^* \rangle u_1^* = x^2 - \frac{650}{12} - \frac{1859}{650} \left(x - \frac{78}{12} \right), \quad (14)$$

$$u_2^* = \frac{v_2}{\|v_2\|_2} =; \quad (15)$$

$$(16)$$

b

Set, $\hat{\varphi} = \sum_{n=0}^2 a_n \cdot x_n$, where a_i are the coefficients to be determined by the equation below.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} \langle y(x), u_0^* \rangle \\ \langle y(x), u_1^* \rangle \\ \langle y(x), u_2^* \rangle \end{pmatrix}$$

c

Orthonormal basis can be reused ,the coefficient matrix of the orthonormal basis is unit matrix, it can be reused.

V

Theorem 5.66

Prove: Because A^+ is given by $A^+y = \sum_{j=1}^r \frac{1}{\sigma_j} \langle y, v_j \rangle u_j$ expressed in matrix form, we have $A^+ = U\Sigma^+V^T$, and $A = V\Sigma U^T$.

PDI-1: consider $A = (a_1, a_2, \dots, a_n)$

$$AA^+a_i = A \cdot \sum_{j=1}^r \frac{1}{\sigma_j} \langle a_i, v_j \rangle u_j \quad (17)$$

$$= \sum_{j=1}^r \frac{1}{\sigma_j} \langle a_i, v_j \rangle \cdot Au_j \quad (18)$$

$$= \sum_{j=1}^r \frac{1}{\sigma_j} \langle a_i, v_j \rangle \cdot \left(\sum_{k=1}^r \sigma_k \langle u_j, u_k \rangle v_k \right) \quad (19)$$

$$= \sum_{j=1}^r \frac{1}{\sigma_j} v_j^T a_i \cdot \sigma_j v_j \quad (20)$$

$$= \sum_{j=1}^r v_j v_j^T a_i \quad (21)$$

$$= a_i \quad (22)$$

So, $AA^+A = A$.

PDI-2 is as same as PDI-1.

PDI-3: Because U and V are orthonormal matrix, we have $U^TU = V^TV = I$, and $\Sigma^+\Sigma = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ Therefore,
 $AA^+ = V\Sigma^+U^TU\Sigma V^T = V \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} V^T$ it is obvious that $(AA^+)^* = AA^+, (A^+A)^* = A^+A$ is also true.

□

Theorem 5.67

Prove:

If A has linearly independent columns, $A = (a_1, \dots, a_n)$, then $A^*A = (a_1^*a_1 + \dots + a_n^*a_n)$ is full rank, so A^*A is invertible.

Consider $A = V\Sigma U^*$, $A^* = U\Sigma^*V^*$, set A has singular value as $\delta_1, \dots, \delta_r$,

so $A^*A = V\Sigma^*U^*U\Sigma V^* = \text{diag}\delta_1^2, \dots, \delta_r^2$.

Therefore, $(A^*A)^{-1}A^* = U\Sigma^+V^T = A^+$

also it is obvious that $A^+A = I$

Similarly, AA^+ is true when A has linearly independent rows.

□

Acknowledgement

Give your acknowledgements here(if any).